Geometric Constraints on Transition Dynamics in Recursive Language Systems

Phase-boundary localization, curvature modulation, and phase-sensitive observables

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Abstract

Transition dynamics in recursive language systems are geometrically structured. States embedded in a Fisher-geometric manifold exhibit transitions concentrated near a phase boundary, with local curvature modulating transition likelihood.

We introduce the Lazarus score, a scalar observable tracking compression along trajectories. Although Lazarus correlates with boundary distance, it retains predictive power within fixed geometric strata: near the boundary, high-Lazarus states show substantially higher transition probability than distance-matched low-Lazarus states, whereas this separation vanishes outside the transition region.

These properties persist across probe scales from 10^1 to 10^5. Transition dynamics are therefore constrained by Fisher geometry, with effective observables arising as projections of that structure.

1. Introduction

Transition dynamics in recursive language systems are often attributed to token statistics, optimization artifacts, or stochastic drift, particularly in the context of recursive generation and model collapse [6–8]. We instead model states as points on a Fisher-geometric manifold [1–4] and show that transitions concentrate near a phase boundary with elevated curvature. This accounts for both localization and reproducibility.

We further introduce the Lazarus score and show that, while correlated with boundary proximity, it remains predictive within fixed geometric strata, resolving structure not captured by distance alone. Transition dynamics are thus governed by manifold geometry, with effective observables arising as projections of that structure.

2. Methods

State space. We construct a state space over recursive outputs using parameterized probes indexed by control parameters *(r, α)* and associated observables.

Fisher geometry. We compute the Fisher information metric to define a Riemannian manifold [1–4]. Distances between states are derived from this metric.

Embedding and trajectories. A weighted graph built from pairwise Fisher distances is embedded for visualization with multidimensional scaling (MDS) [5], while graph geodesics approximate trajectories through the manifold.

Phase boundary and Lazarus. We identify a boundary separating stable and unstable regimes using curvature and transition signals. The Lazarus score aggregates compression along probe paths as a scalar summary of transition readiness, with higher values indicating stronger trajectory-level compression.

Transitions. Transitions are defined as regime changes within a fixed horizon *k* along a trajectory and represented as a binary variable.

AI use statement. A large language model was used to assist with language development, structural editing, figure-caption drafting, and manuscript refinement. All scientific decisions, analyses, interpretations, and final wording were reviewed and approved by the author, who takes full responsibility for the content of the manuscript.

3. Results

States organized by distance to the phase boundary and local curvature separate into three regimes: a high-curvature boundary, an intermediate transition zone, and a low-curvature interior. Transition probability concentrates near the boundary and yields a transition ridge that decays with distance.

Figure 1 — Geometric structure of the state space

Transitions concentrate near the phase boundary along a geometric ridge

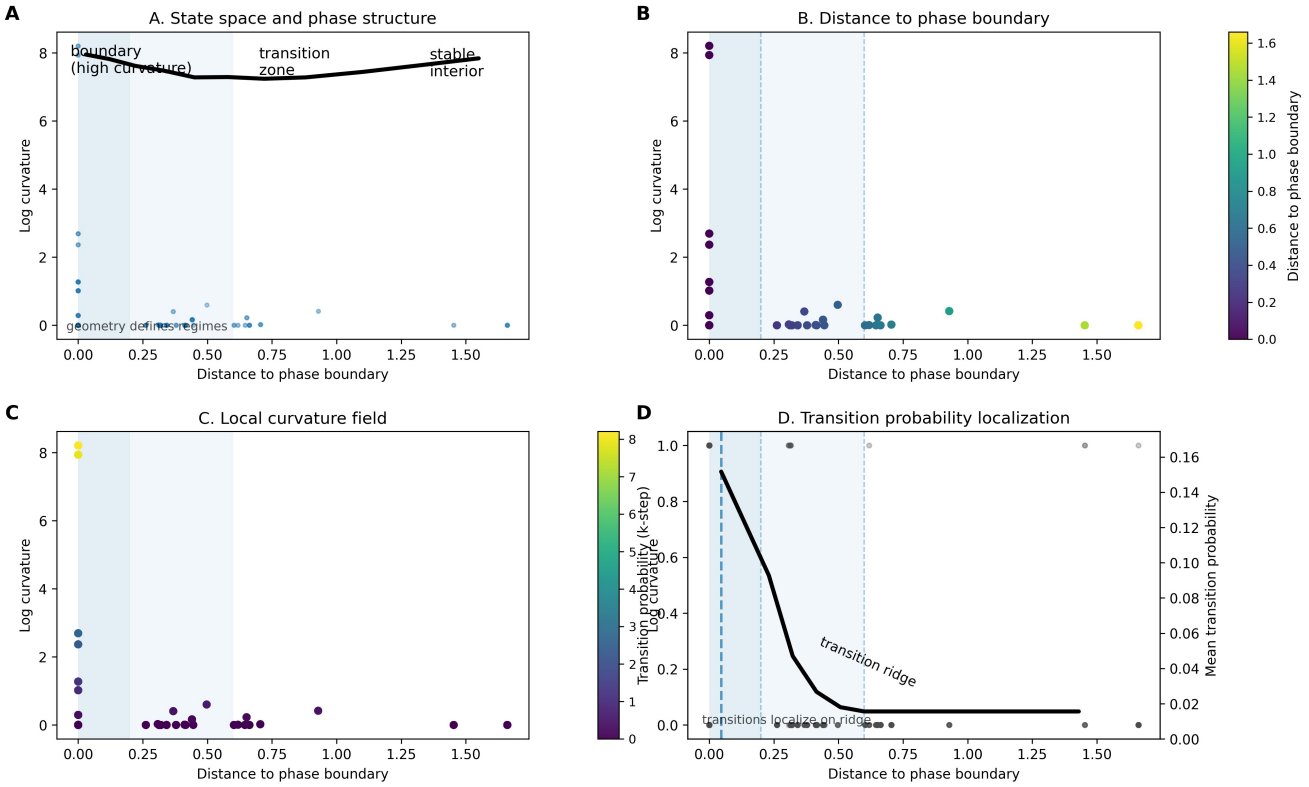


Figure 1 — Geometric structure of the state space. (A) States in distance–curvature space show a boundary, transition zone, and interior. (B) Distance reveals sharp stratification. (C) Curvature localizes near the boundary. (D) Transition probability peaks near the boundary and aligns with the transition ridge.

High-Lazarus states show substantially higher transition probability than low-Lazarus states. Compression precedes phase change and is enriched near the boundary, linking a low-dimensional observable to geometric structure.

Figure 2 — Transition-rate law

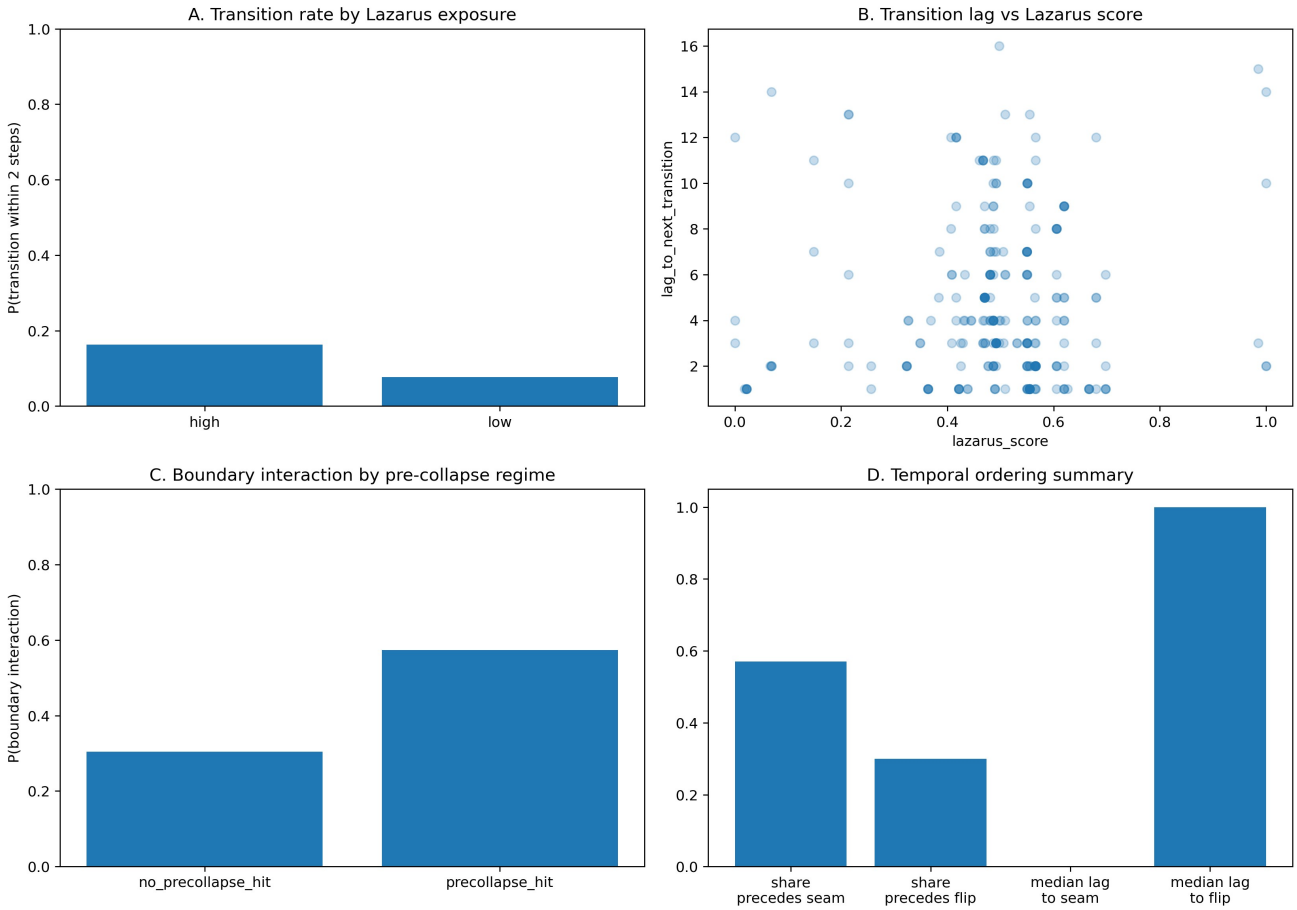


Figure 2 — Transition-rate law. High-Lazarus states show substantially higher transition probability than low-Lazarus states. Temporal ordering and boundary-interaction summaries indicate that compression precedes phase change and is enriched near the boundary.

Conditioning on distance and curvature reveals a sharp separation: distance acts as a gate, while curvature modulates within this constraint. Transition probability concentrates on a low-dimensional manifold and is suppressed away from it.

Figure 3 — Transition mechanism

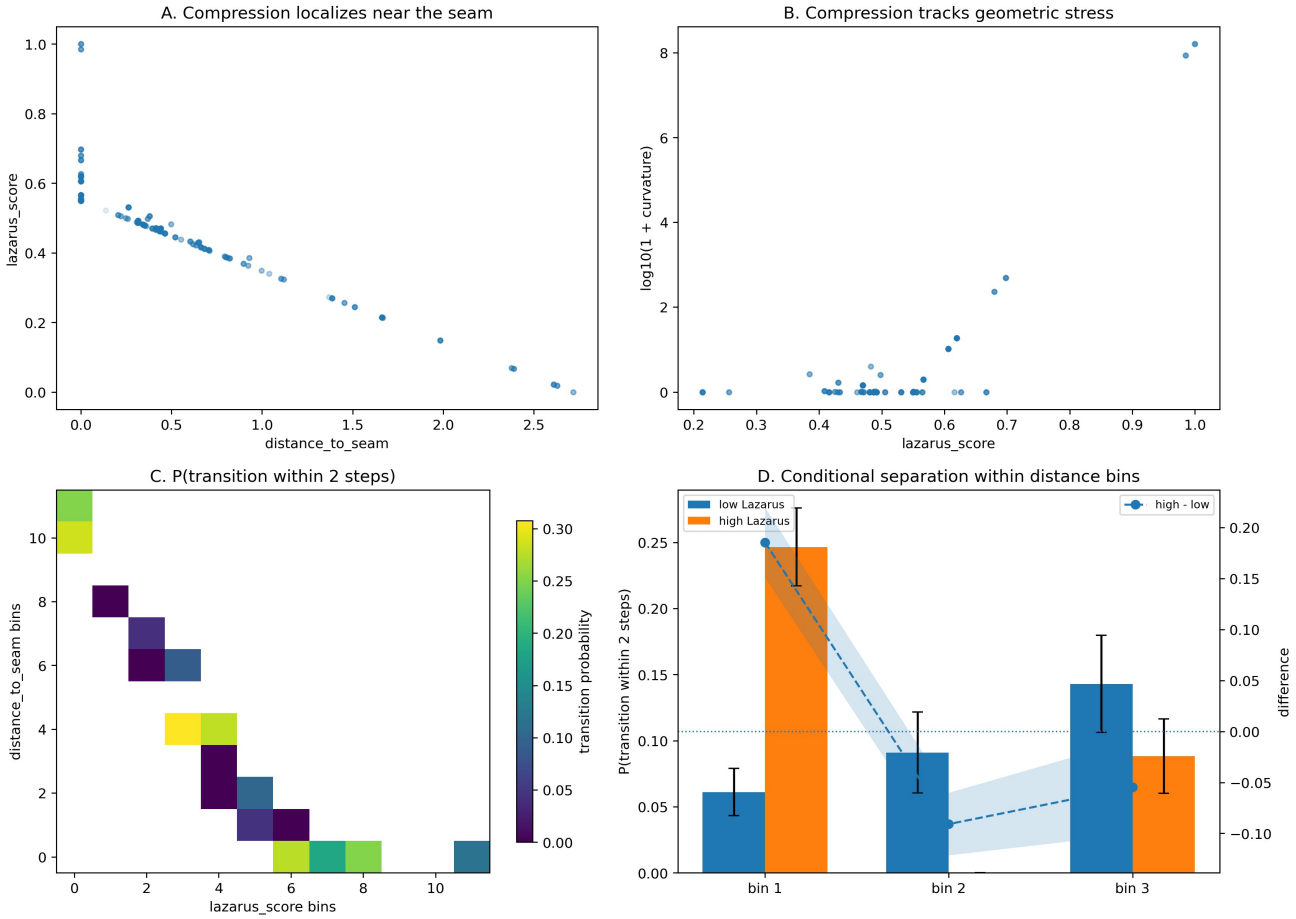


Figure 3 — Transition mechanism. Compression localizes near the phase boundary, tracks geometric stress, and exhibits conditional separation within distance bins. Near-boundary high-Lazarus states are most transition-prone, while this separation weakens away from the critical region.

The observed law is stable across probe scales from 10 to 100,000. Transition-rate differences, lag statistics, and boundary-interaction measures converge, indicating that the effect is not a small-sample artifact.

Figure 4 — Scaling robustness

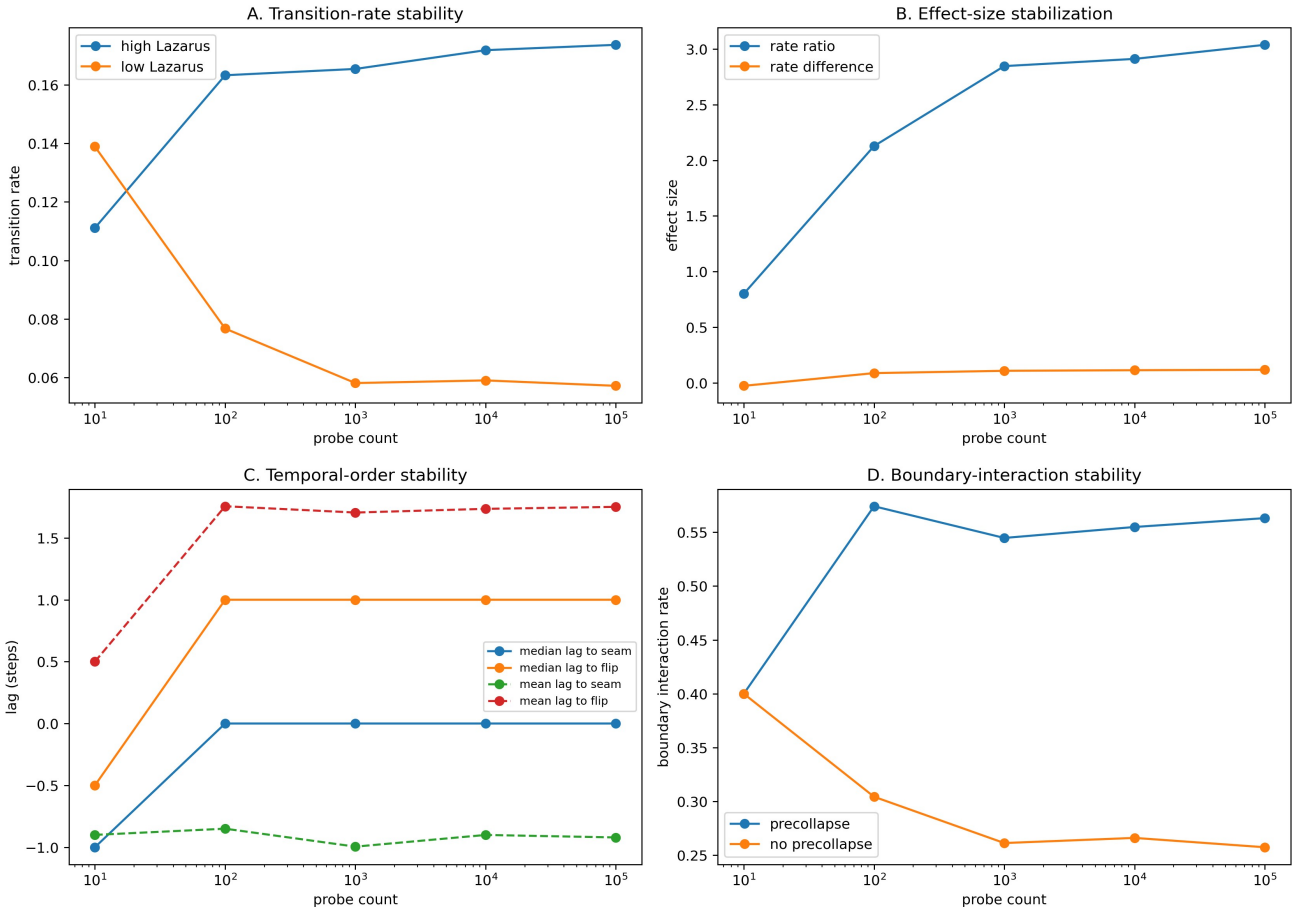


Figure 4 — Scaling robustness. Transition-rate differences, lag statistics, and boundary-interaction measures stabilize across probe scales from 10 to 100,000, indicating that the observed law is not a small-sample artifact.

4. Discussion

Transitions are localized by geometry rather than uniformly distributed. Boundary distance controls access to transitions: interior states remain stable, while boundary states exhibit elevated rates. Curvature modulates within this gate but does not independently induce transitions. Distance controls access; curvature shapes expression.

No single scalar governs the dynamics. Scaling appears only within distance-defined regimes, indicating piecewise structure. The state space behaves as a curved manifold [2,3], with distance transverse to a transition surface, and the transition ridge marks elevated susceptibility.

Instability in iterative systems is spatially organized rather than purely stochastic, complementing recent concerns about collapse in recursively trained generative systems [6–8]. Control may therefore be achieved by maintaining distance from critical boundaries, enabling prediction from geometric diagnostics. Results depend on the chosen embedding and metric, and transition estimates remain finite-horizon, but the primary observables are computed before visualization and remain stable across scale.

5. Conclusion

Transition dynamics in recursive language systems are constrained by Fisher geometry: they localize near a phase boundary and are modulated by curvature. The Lazarus score provides a low-dimensional observable that resolves structure within this manifold, especially in the critical regime.

Predictive observables for regime change can thus emerge directly from geometry induced by system dynamics.

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